

Who Counts the Trials?

A Committed Trial Ledger for Enforcing the Deflated Sharpe Ratio in Zero-Knowledge

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ABSTRACT

The Deflated Sharpe Ratio (Bailey and López de Prado, 2014) corrects an observed Sharpe ratio for the number of trials N behind it, separating genuine skill from the selection bias of a large backtest search. Its practical weakness is structural: N is supplied by the same researcher whose result it constrains. A search over a thousand configurations, reported as a single trial, satisfies the formula while defeating its purpose. The correction is sound; its input is self-reported.

We present a construction that removes the researcher's discretion over that input. The trial set is committed to a Merkle tree before evaluation; the trial count N is the **leaf count of the tree**, not a reported scalar; and the winning strategy is bound, inside a zero-knowledge proof (a STARK), to be the **maximum over the committed leaves**. The deflation is then recomputed *in-circuit* on Merkle-pinned prices, net of a cost model the credential discloses, so the figure an allocator reads is derived by the circuit rather than asserted by the manager. The output is a credential, checkable by anyone, in seconds, without disclosure of the strategy, whose anti-overfitting correction **cannot be understated within the committed search**.

We give the commitment scheme and its in-circuit binding; state precisely the manipulation it eliminates (understating N , cherry-picking a non-maximal winner, softening the cost model) and the residual trust it does not (off-ledger trials, closed only by forward pre-registration); and report a live implementation that additionally computes, in-circuit, the Probability of Backtest Overfitting over all $C(16,8) = 12,870$ combinatorially-symmetric splits (via recursive proof composition), together with the Probabilistic Sharpe Ratio and Hansen's Superior Predictive Ability. We demonstrate the system on its own flagship strategy, which it rules **not significant** (DSR 0.68, below the 0.95 bar), and publish that failure as the reference credential.

KEYWORDS: deflated Sharpe ratio, backtest overfitting, multiple testing, zero-knowledge proof, STARK, verifiable computation, selection bias, systematic trading. **JEL:** C12, C52, G11, G17.

1. Introduction

A systematic track record is, almost universally, self-reported. A manager runs a search over strategies, selects the one that performed best, and presents its Sharpe ratio, its drawdown, and a narrative of the process. The allocator on the other side must decide whether the number reflects a genuine edge or the survivorship of a wide search, with essentially nothing to check it against but trust. The reported figure,

the number of configurations tried, the costs assumed, even the data the strategy was scored on, are all supplied by the party whose result they judge.

This is not a marginal problem. The literature has established, repeatedly, that a sufficiently large backtest search produces impressive in-sample performance from no skill whatsoever (Bailey, Borwein, López de Prado, and Zhu, 2017; Harvey and Liu, 2020), and that the correction for this selection bias depends on inputs, the trial count above all, that the searcher alone knows and can understate. The field has excellent mathematics for separating skill from luck. What it lacks is any mechanism to compel that the mathematics be applied to honest inputs.

Zero-knowledge proof systems supply that mechanism, along the one axis that dissolves the usual objection. The reason a manager will not submit to independent verification is that the strategy is the asset: to reveal it is to destroy it. A zero-knowledge proof lets a computation be checked without its inputs being disclosed. Applied to a backtest, it makes verification and secrecy compatible for the first time: a manager can prove that a track record survives the field's own anti-overfitting tests without revealing the strategy that produced it.

This paper describes a construction that does so, and reports a live implementation. Our contributions are:

- **A committed trial ledger** that removes the searcher's discretion over the deflation's inputs: the trial count is the leaf count of a committed Merkle tree, and the presented winner is bound in-circuit to be the maximum over the committed trials (§3).
- **In-circuit computation of both schools** of the anti-overfitting literature, the in-sample deflation (Deflated and Probabilistic Sharpe, PBO, CPCV) and the multiple-testing correction (Hansen's SPA), in integer-faithful arithmetic identical on the host and in the proof. The Deflated Sharpe and composed-PBO paths are stamped today; the Probabilistic Sharpe and SPA compute in-circuit in the current build and are being finalized into a stamped era (§4.3).
- **A sealed-model path** that lets any strategy, machine learning, code, or discretion, be proven from its committed decisions alone, so the logic never leaves the manager's machine (§5).
- **A self-failing reference credential:** we demonstrate the system on our own flagship strategy, which it rules not significant, and publish that verdict (§6). A verification layer that cannot fail its own author is not credible.

We are equally precise about what the construction does not establish. A sealed backtest, however honest its arithmetic, remains self-reported history; the trial ledger binds the committed search but not the existence of trials run outside it; and the no-lookahead guarantee is earned only forward, by committing positions before the bars they trade exist. We treat these boundaries as first-class and return to them in §7. The sections that follow make each contribution precise, and, in §7, equally precise about its limits.

2. Background and Prior Work

2.1 *Correcting a backtest for selection bias*

Two overlapping literatures confront the same failure: a strategy chosen as the best of many looks skilled even when none of the candidates is. The first, *in-sample deflation*, adjusts a performance statistic downward for the breadth of the search that produced it. The Probabilistic Sharpe Ratio (Bailey and López de Prado, 2012) expresses the confidence that a track record's true Sharpe exceeds a threshold given its length, skew, and kurtosis; the Deflated Sharpe Ratio (Bailey and López de Prado, 2014) extends this by deflating the observed maximum for the number of trials N and their dispersion; the Probability of Backtest Overfitting (Bailey, Borwein, López de Prado, and Zhu, 2017), computed by combinatorially-symmetric cross-validation (CSCV) and later combinatorial purged cross-validation (López de Prado, 2018), estimates how often the in-sample best underperforms out-of-sample median.

The second literature, *multiple-testing correction for data snooping*, controls the family-wise error of searching many strategies at once: White's Reality Check (2000), Hansen's test for Superior Predictive Ability (2005), and the stepwise procedures of Romano and Wolf (2005), carried into asset pricing by Harvey, Liu, and Zhu (2016) and Harvey and Liu (2020).

For all their differences, these methods share one structural dependency: **each is parameterized by the size of the search (the trial count, the family of tests, the strategy universe), and that quantity is supplied by the researcher.** The correction that exists to police selection bias takes its most important input from the party being policed. Worse, even an honest researcher frequently cannot reconstruct the true search, because a realistic modeling process branches at every data-cleaning and specification choice (the "garden of forking paths," Gelman and Loken, 2013). The trial count is thus both *gameable* and often genuinely *unknown*, and no amount of statistical refinement to the formula addresses either, because the problem is not in the mathematics but in the absence of a mechanism that binds its inputs.

2.2 Zero-knowledge proofs and verifiable computation

A zero-knowledge proof (Goldwasser, Micali, and Rackoff, 1985) lets a prover convince a verifier that a statement is true while revealing nothing beyond its truth. Modern succinct, transparent proof systems, in particular STARKs (Ben-Sasson, Bentov, Horesh, and Riabzev, 2018), which need no trusted setup and are publicly verifiable, make it practical to prove that a general computation was executed correctly on committed inputs. Executed inside a zero-knowledge virtual machine, an arbitrary program becomes a statement one can prove without disclosing its private data. Combined with Merkle commitments (Merkle, 1988), which bind a body of data to a single root that cannot later be altered, this yields exactly the primitive systematic finance needs: *prove that a strategy cleared a defined statistical gate, on fixed data, without revealing the strategy.*

2.3 Verifiable claims in finance

Cryptographic verification has entered finance, but along axes orthogonal to statistical honesty.

- **Proof of solvency and reserves.** Provisions (Dagher, Bünz, Bonneau, Clark, and Boneh, 2015) and its successors let an exchange prove its assets exceed its liabilities without revealing balances. This proves a *custodial* property, that value exists, a static fact about a balance sheet, not the validity of a performance claim.

- **Timestamped provenance.** Commit-and-timestamp services (e.g. vBase) and Bitcoin-anchored timestamping (OpenTimestamps) establish that a dataset or prediction existed at a given time. This proves *when*, which is necessary for a no-lookahead forward record but says nothing about whether a reported edge survives selection bias.
- **Verifiable machine-learning inference.** zkML systems (e.g. Kang, Hashimoto, Stoica, and Sun, 2022) prove that a specific model produced a specific output. This proves a *computation ran*, not that the resulting track record is statistically sound.
- **Emerging ZK trading and credentialing.** Several commercial efforts apply zero-knowledge proofs to execution provenance or headline performance attestation. To our knowledge, none enforce the anti-overfitting framework, the deflation of a result for the breadth of the search behind it.

2.4 The gap this paper addresses

Each strand above proves something real and something adjacent: that assets exist, that data predates an event, that a model executed, that a trade occurred. None proves the property an allocator actually needs before trusting a systematic track record: that the *number itself is honest* once corrected for how hard it was searched for. Delivering that requires placing the correction's own inputs — the trial count above all, but also the winner, the dispersion, and the disclosed cost model — beyond the discretion of the party reporting them. That is the contribution developed in §3 and extended to the full statistical suite in §4. To our knowledge, this is the first construction to enforce the anti-overfitting framework of systematic finance inside a zero-knowledge proof.

3. The Committed Trial Ledger

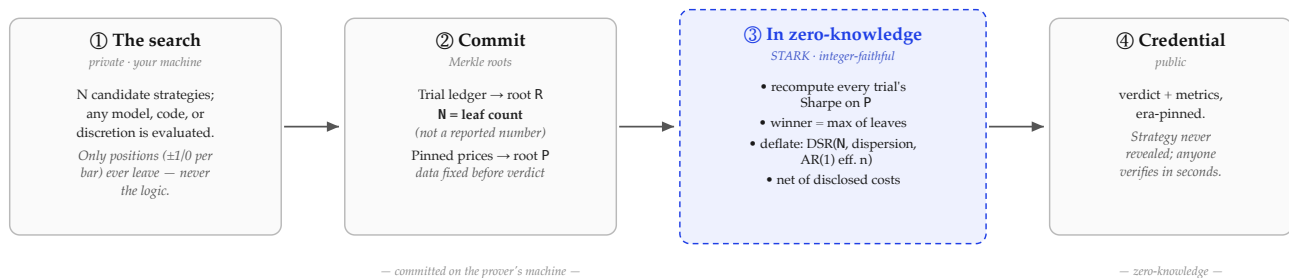


Figure 1 — The construction. A search of N strategies is committed to a Merkle ledger (its size, N , is the leaf count) against pinned prices; a zero-knowledge proof recomputes each trial's Sharpe, binds the winner to be the maximum of the committed leaves, and deflates the result — emitting a public credential that reveals the verdict but never the strategy.

3.1 The problem: a self-reported denominator

Let a backtest search produce trials $t_1, \dots, t_N$, each a strategy configuration evaluated on a common price history, yielding a realized Sharpe estimate $\hat{s}_i$. The researcher reports the best, $\hat{s}^* = \max_i \hat{s}_i$. The Deflated Sharpe Ratio asks whether $\hat{s}^*$ exceeds what the *best of N independent lucky draws* would produce under the null of zero true skill, using the expected maximum

$$E[\max] \approx \sqrt{\text{Var}(\hat{s}_i)} \cdot [(1 - \gamma) \cdot Z^{-1}(1 - 1/N) + \gamma \cdot Z^{-1}(1 - 1/(N \cdot e))]$$

where γ is the Euler–Mascheroni constant and Z^{-1} the inverse standard normal. The correction therefore consumes three researcher-supplied quantities: the **winner** $\hat{s}^*$, the **trial count** N , and the **dispersion** $\text{Var}(\hat{s}_i)$ across trials.

Every one of the three is, in current practice, asserted. Nothing binds the reported N to the search actually run; nothing forces the presented result to be the true maximum of that search (a researcher may present a *typical* configuration and quietly quote a low N); nothing prevents $\text{Var}(\hat{s}_i)$ from being narrowed by omitting the diffuse tails of the search. The statistic that exists to police selection bias is, in deployment, parameterized by the party being policed. This is not a flaw in the mathematics; it is the absence of an enforcement layer beneath it.

3.2 Construction: committing the trial set

Before any deflation is computed, the researcher (or their machine, offline) constructs a **trial ledger** (Figure 1): a Merkle tree whose leaves are commitments to the individual trials of the search.

- **Leaf.** Leaf i commits to trial t_i : at minimum a binding hash of its per-bar position stream $\pi_i \in \{-1, 0, +1\}^T$ (fractional positions permitted), together with the identifier of the price series it was evaluated on. In the full-soundness variant (§3.4) the leaf commits the position stream itself; in the light variant it commits the stream and its attested Sharpe.
- **Ledger root.** The Merkle root R is the single value published in the credential. It fixes the entire search, its membership, its size, and the content of every trial, before any outcome is revealed.
- **Trial count.** N is defined as the **number of leaves under R** , proven from the tree's structure, not a field the researcher fills in. To claim a smaller N , the researcher would have to exhibit a different tree with a different root, a publicly distinct credential.

The price data itself is pinned the same way: the evaluation prices are committed to a separate Merkle structure and referenced by root, so "the data the strategy was scored on" is fixed and cannot be swapped after the fact.

3.3 In-circuit binding: N is the count, the winner is the maximum

The zero-knowledge circuit takes the ledger root R , the pinned-price root P , and a claimed winning trial, and proves, without revealing any strategy, the following statements jointly:

- **Membership.** The claimed winner is a leaf under R (standard Merkle inclusion).
- **Faithful recomputation.** Re-running the winner's committed position stream on the pinned prices P , under a fixed cost model, reproduces the claimed winning Sharpe $\hat{s}^*$. The number is *derived inside the proof*, integer-faithful down to the truncating division and fixed-point equity the circuit uses; it is not read from the leaf.

- **Maximality.** $\mathcal{S}^*$ is $\geq$ the Sharpe of every other leaf under R , each recomputed in-circuit on the pinned prices at a canonical unit leverage, so the winner and every leaf are ranked on the same per-observation basis (annualization is a constant scale that preserves the ranking). No committed trial scored higher than the one presented. This is the statement that forbids presenting a non-maximal result while hiding the true winner elsewhere in the search.
- **Deflation.** N (the leaf count), the dispersion over the in-circuit-recomputed trial Sharpes, the winner's own length and higher moments (skew, kurtosis, and an AR(1) correction to the effective sample size), and the disclosed trading costs are combined into the Deflated Sharpe Ratio *by the circuit*. The fee and slippage are committed by the prover, applied in-circuit to the equity curve, and recorded in the credential; because those committed costs are public, a verifier can reject any credential whose disclosed costs fall below a required protocol floor. The credential carries the result and every input it was computed from.

The verifier — an allocator, a counterparty, anyone — checks the STARK in seconds and learns exactly one thing beyond the verdict: that the deflation was computed correctly, on a fixed search of a fixed size, against a fixed dataset, net of a disclosed cost model a conforming verifier can reject if it falls below a required floor. They learn nothing about the strategy.

3.4 What this eliminates

Relative to a self-reported DSR, the committed ledger closes two manipulations by construction, the trial count and the maximal winner, and a third, a softened cost model, by a public disclosure a conforming verifier can reject:

- **Understating N .** N is the leaf count of the committed tree. A search of a thousand configurations cannot be reported as ten without publishing a ten-leaf tree that commits to only those ten, at which point the other 990 were, by the researcher's own commitment, *not part of this credential's search*, and the claim stands or falls on that public fact rather than on their word.
- **Cherry-picking a non-maximal winner.** The maximality clause binds the presented result to be the best of the committed set. One cannot commit a wide search and then quote a modest, "representative" member while the true maximum sits unmentioned in the same tree.
- **Hiding the cost model.** The costs are committed, applied in-circuit, and recorded in the credential. A frictionless or optimistic-fill assumption cannot be concealed: the reported Sharpe is provably net of the costs the credential discloses, and a verifier can reject any credential whose disclosed costs fall below a required floor.

SOUNDNESS VARIANTS

In the *full-soundness* variant the circuit recomputes every leaf's Sharpe from its committed positions on the pinned prices, so the dispersion term and the maximality check rest entirely on in-circuit computation; proving cost scales with the search size. In the *light* variant only the winner is recomputed and the remaining leaves contribute attested Sharpes bound by commitment; this is cheaper and still binds N and the winner's own derivation, at the price of trusting the committed (not recomputed) Sharpes of the losers

for the dispersion term. We regard the full-soundness variant as the standard for a credential of record and the light variant as an optimization for large searches; the credential states which was used.

3.5 What it does not eliminate: off-ledger trials

The ledger binds the search *the researcher chose to commit*. It cannot, on its own, prove that this search is the *entirety* of what the researcher ever ran. A determined manipulator may run a large private search off-ledger, then commit only a small, favorable subtree and present the best of it with an honestly-counted small N . The construction converts the trust assumption from "*trust the reported number N* " into "*trust that the committed ledger is the whole search*" — strictly weaker, and internally consistent where the old one was not, but not zero.

We state this plainly because the boundary is the point. The residual gap has a single principled closure: **the trials must be committed before their outcomes can be known**. Under forward pre-registration — an anchored ledger whose root is timestamped (e.g. to a public blockchain) before the out-of-sample bars it will trade exist — off-ledger re-searching gains the manipulator nothing, because a trial not committed in advance cannot be substituted in after the outcome is seen without breaking the anchor. The committed ledger of this section is the in-sample enforcement layer; forward anchoring is its out-of-sample completion. A credential that carries only the former is, and is labeled as, a *sealed backtest*: self-reported history whose arithmetic is now honest, but history nonetheless.

3.6 Security properties (informal)

- **Completeness.** An honest researcher with a genuine best-of- N result can always produce an accepting proof.
- **Soundness.** In the full-soundness variant (the shipped guest path), except with negligible probability an accepting proof implies: the winner is a member of the committed tree of size N ; its Sharpe equals the in-circuit recomputation on the pinned prices net of the committed cost model; and no committed leaf exceeds it. Understating N , presenting a non-maximal winner, and reporting a Sharpe not net of the disclosed costs are therefore infeasible *within the committed search*. In the light variant the maximality and dispersion terms rest on the committed (attested) loser Sharpes, and are sound only under that commitment.
- **Zero-knowledge.** The proof reveals nothing about any strategy beyond the public statements above; in particular the position streams, the logic that generated them, and the identities of the non-winning trials remain hidden.
- **Binding across time.** With a timestamped root (§3.5), the search is fixed relative to a wall-clock anchor, the precondition for extending an in-sample credential into a no-lookahead forward track. (This is a *roadmap* property: the anchor-predates-data rule is not yet enforced in-circuit, see §7.)

4. Computing the Suite In-Circuit

The committed trial ledger of §3 fixes the *inputs* to the deflation. This section describes how the statistics themselves are computed inside the proof, and states honestly which are part of a stamped credential

and which are computed but still being finalized into a released engine era.

4.1 *Integer-faithful arithmetic*

Every statistic in the proof is computed in fixed-point integer arithmetic, at a scale of 10^6 , with a fixed truncating division order. There is no floating point anywhere on the proven path. This is not an implementation detail but a soundness requirement: floating-point results depend on evaluation order and hardware, and a proof whose output could differ between the prover's machine and a verifier's recomputation would be worthless. The circuit's arithmetic is bit-identical on the host and in the RISC-V guest, so any party can reproduce a committed number exactly (§6.4 demonstrates this). Floating-point reference implementations of the statistics exist for host-side cross-checking, but are never on the proven path.

4.2 *The in-sample school*

The Deflated and Probabilistic Sharpe Ratios (§3), the Probability of Backtest Overfitting, and combinatorial purged cross-validation are computed in-circuit. The PBO is evaluated by combinatorially-symmetric cross-validation over the full set of $C(16,8) = 12,870$ balanced splits. Because a single proof over all 12,870 splits is impractically large, the computation is structured as *recursive proof composition*: subproofs are verified inside an aggregating proof, so the final credential attests the complete split set without a monolithic circuit. CPCV is computed with canonical purge and embargo counts enforced in-circuit.

4.3 *The multiple-testing school*

Hansen's test for Superior Predictive Ability is computed in-circuit by a circular block bootstrap on committed returns with a committed seed, so the bootstrap is deterministic and reproducible by any verifier. The minimum backtest length implied by the reported Sharpe is computed as a consistency check; we note honestly that this check is applied verifier-side against the committed inputs rather than fully re-derived within the circuit.

These extended validation statistics compute in-circuit in the current engine build and are being finalized into the stamped credential era; the Deflated Sharpe and composed-PBO figures reported in §6 are drawn from stamped credentials. We distinguish the two explicitly rather than assert a uniform maturity the implementation does not yet have.

4.4 *Era pinning*

Each credential names the exact engine image that produced it, by cryptographic identifier, and verifies against that engine permanently. A credential minted under one engine era remains checkable against that era forever, and is never silently re-labeled when the engine advances. This fixes a credential's meaning precisely — *this result, under this exact verifier, on this committed data* — and makes an engine upgrade a new era rather than a retroactive change to past credentials.

5. Proving Without Revealing: The Sealed Model

A proof of honesty is worthless to a manager if producing it requires disclosing the strategy. Every credential in this construction is therefore zero-knowledge to its reader: an allocator who verifies it learns the verdict and the committed metrics and nothing about how the positions were generated. What differs between submission paths is not what the world sees, but what the *prover* sees.

In the *language path*, the strategy is expressed in a constrained domain-specific form that the prover evaluates to derive positions. In the *sealed path*, the prover never sees the logic at all. The strategy — a neural network, a C++ stack, a discretionary process — runs on the manager's own hardware, and what is submitted is only the decision stream: one position per bar, long, short, or flat, fractional permitted. The circuit compounds that stream on the pinned prices, net of the disclosed costs, and computes the same statistics. The logic never leaves the machine.

The commitment is what gives the seal its force. The decision stream is bound to a cryptographic commitment before the verdict is known. Altering a single decision after the fact changes the committed root, and the credential then fails every verifier. One cannot seal a strategy, observe a poor result, and quietly substitute a better one: the object judged is provably the object committed.

The sealed path is the maximum-privacy end of the spectrum, and it carries honest limits. A commitment establishes immutability, not the absence of hindsight: a manager who constructs a position stream with lookahead can commit and prove it, and the seal will faithfully certify a lookahead-contaminated backtest. Positions are also a lower-fidelity disclosure than logic; an allocator learns less about the source of the edge. And the circuit proves that the committed positions, on the committed prices, produce the stated metrics; it does not, and cannot, prove that those positions were generated by any particular model, or that the prices are the true market. These are the boundaries of §7, which the sealed model does not escape; it is the reason the forward-anchored track of §7 exists.

6. Worked Example: A Self-Failing Credential

Theory is inexpensive, and a paper about verification should verify something. We therefore demonstrate the construction on a live credential in the MIZAN registry, and, deliberately, on one that *fails*. The reference strategy is our own: a long-only BTC trend book, evaluated on daily bars. We choose a failing credential of our own making because a verification layer that only ever displays its successes proves nothing about its honesty, and because every input below can be reproduced by the reader without trusting our software.

6.1 Committed inputs

The credential fixes, before any verdict, a committed trial ledger of $N = 10$ configurations and a pinned daily price series. The winning configuration's realized statistics, recomputed in-circuit on the pinned prices net of the committed cost model (fee and slippage, disclosed in the credential), are:

Quantity	Value
In-sample Sharpe	0.75
Out-of-sample Sharpe	1.10

Maximum drawdown	16.16%
Net return	47.6%
Compound annual growth	13.9%
Worst single bar	-13.54%
Committed trial count N	10

None of these is asserted by the manager: each is derived by the circuit from the committed positions on the pinned prices, integer-faithful to the truncating division and fixed-point equity the circuit uses.

6.2 The deflation

Applying the Deflated Sharpe Ratio with $N = 10$ and an AR(1)-corrected effective sample size, on the committed dispersion across the ten trials:

$$\text{DSR} = 0.6779 \rightarrow \text{verdict: NOT significant (below the 0.95 bar)}.$$

An in-sample Sharpe of 0.75, once deflated for the breadth of the search that found it, does not reach 95% confidence of positive true skill. The credential records this verdict and is published unchanged. A strategy that does not clear its own deflation is not dressed up; it is recorded honestly, and its failing credential sits in the same public registry, under the same locked engine, as every passing one.

The same engine clears strategies that survive their search. A second strategy of our own, sealed rather than disclosed, records a raw in-sample Sharpe of 1.34 over $N = 9$ committed trials, which deflates to a Deflated Sharpe of 0.96, clearing the 0.95 bar (significant at 95%). The contrast is the point: an undeflated Sharpe of 0.75 fails, while a 1.34 on a comparably-sized search clears. The deflation is not a rubber stamp in either direction; it separates a marginal result from a real one, and the same locked engine issues both verdicts, on our own strategies, and we publish both. Only the failing strategy is disclosed for reproduction (§6.4); the passing one is sealed, like every credential an allocator would actually submit.

6.3 Companion statistics

The same engine composes the rest of the suite over the identical committed inputs:

Statistic	Value	Reading
Deflated Sharpe Ratio	0.6779	not significant (< 0.95)
Probability of Backtest Overfitting	0.0759	not overfit
CPCV, 5th-percentile path Sharpe	+0.003	barely positive under adverse splits

The Deflated Sharpe and PBO figures are drawn from stamped v10 credentials; CPCV is computed in-circuit. The multiple-testing school (PSR, MinBTL, Harvey-Liu, SPA) runs on the current v11 build and is shown separately below.

Two of these deserve a careful, honest reading, because they do not say the same thing. The PBO of 0.0759, computed by exact combinatorially-symmetric cross-validation over all $C(16,8) = 12,870$ splits,

indicates the in-sample selection is *unlikely to be an overfit artifact*: it generalizes out-of-sample more often than not. The DSR of 0.6779 indicates the deflated result *does not reach the 95% significance bar*. These are compatible, not contradictory: the strategy is plausibly real but not proven significant at the chosen confidence. The engine reports both and lets neither hide the other, which is precisely the discipline a self-reporting manager is free to abandon, quoting whichever statistic flatters the book.

That the PBO test has teeth rather than blessing everything is itself demonstrable: run the identical 12,870-split procedure on constructed scenarios where the truth is known, and a pure-noise search is flagged (PBO $\approx$ 0.44) while a genuine-skill search clears (PBO $\approx$ 0.001). The flagship's 0.0759 sits squarely in the skill regime.

Both schools, on one credential. The example above carries the in-sample school (DSR, PBO, CPCV) on a stamped v10 credential. The current v11 build additionally computes the multiple-testing school in-circuit, over a single committed ledger. On a representative credential of our own ($N = 9$ committed trials of roughly 12,900 bars), the four extended corrections return:

Correction	Value	Verdict
Probabilistic Sharpe Ratio	0.7226	below the 0.95 bar $\rightarrow \times$
Minimum backtest length	12,916 / 2,438	have $\gg$ required $\rightarrow \checkmark$
Harvey-Liu (multiple testing)	$t = 3.50$, $p_{\text{BHY}} = 0.012$	below 0.05 $\rightarrow \checkmark$
Hansen SPA	$t_{\text{max}} = 3.49$, $p = 0.005$	below 0.05 $\rightarrow \checkmark$
Composite	—	NOT significant

Three of the four corrections clear; the Probabilistic Sharpe does not; and the honest composite verdict is therefore **not significant**. This is the discipline made concrete: the engine reports every correction and takes the most conservative, rather than quoting whichever one happens to clear. The extended suite computes in-circuit in the current v11 build and is being finalized into a stamped era (§4.3); the proving cost of exactly this credential is the measurement reported in §6.6.

6.4 Reproducibility

Every committed number is reproducible from the pinned dataset alone, with no access to MIZAN's prover. A public recompute script (with the pinned dataset and the disclosed fixture strategy, at mizan.market/research/reproduce) re-implements the in-circuit backtest, integer-faithful, dependency-free, offline, and reproduces each figure to the printed digit:

```
python3 mizan_recompute.py btc_daily.csv
```

metric	engine (committed)	this script	match
Sharpe	0.75	0.75	$\checkmark$
00S Sharpe	1.10	1.10	$\checkmark$
MaxDD %	16.16	16.16	$\checkmark$
Net %	47.6	47.6	$\checkmark$

CAGR %	13.9	13.9	✓
Worst bar %	-13.54	-13.54	✓
Trades	65	65	✓
Bars in mkt	129	129	✓

 EVERY COMMITTED NUMBER REPRODUCED INDEPENDENTLY.

The reader need not trust the verifier to trust the arithmetic, which is the posture of the entire construction, reduced to a single number. The point of the example is not which strategy wins but that one locked engine judges every credential the same way and publishes the failing verdict as readily as the passing one.

6.5 Provenance of the figures

The credential cited here was minted on engine era **v10** (image identifier `dafad185...`) and verifies against the v10 engine permanently: under era law, a credential names the exact engine that judged it and remains checkable against that engine forever. The current engine build (**v11**, `3ac3b10b...`) additionally computes this same Deflated Sharpe alongside the Probabilistic Sharpe Ratio, Hansen's Superior Predictive Ability, and the composed 12,870-split PBO, the two schools of §2 evaluated in-circuit. The DSR and composed-PBO figures above are drawn from stamped credentials in the live registry; the extended validation statistics are computed in-circuit and are being finalized into the stamped era. Every figure above is drawn from a credential in [the live registry](#) and is independently re-verifiable at the time of writing.

6.6 Proving and verification cost

The proofs are not free, and the cost is dominated by re-deriving every committed trial in-circuit rather than trusting attested values. As a representative measurement, the extended validation credential — $N = 9$ committed trials of roughly 12,900 bars each, with all four multiple-testing corrections (the Probabilistic Sharpe Ratio, minimum backtest length, Harvey–Liu, and Hansen's SPA with a 400-resample block bootstrap) computed in-circuit and every trial fully re-derived — proved in about 3,700 s of core proving (roughly 62 minutes, or 74 minutes wall-clock including cold-start) on a single NVIDIA H100 GPU. The resulting proof is on the order of 300 MB and compresses to a **221 KB succinct receipt** that re-verifies in seconds on commodity hardware, with no access to the prover, the data, or the strategy. Proving cost scales with the product of the trial count, the series length, and the number of statistics proven; the single-statistic Deflated Sharpe and composed-PBO credentials are correspondingly lighter. This asymmetry — expensive to prove once, trivial to verify forever — is the standard economics of succinct proofs, and it is the right shape for a credential: the manager bears the cost of proving, and every allocator verifies for free.

7. Limitations

The value of a verification layer is bounded by the honesty of its own boundary. We state ours plainly.

A sealed backtest is self-reported history. The construction proves that a committed strategy, on committed prices, at disclosed costs, cleared or failed the deflation. It does not establish that the positions were free of hindsight. The no-lookahead guarantee is earned only forward, by extending an anchored track — the forward layer this construction defines but does not yet enforce — whose positions are committed before the bars they trade exist. A one-shot sealed proof is honest arithmetic on self-reported history; a growing anchored track is the closest cryptographic analogue of a live record, and the credential an allocator should ultimately weight most.

The ledger binds the committed search, not the entire search. As detailed in §3.5, the trial count is proven equal to the committed ledger's leaf count and the winner to its maximum, but nothing in a single credential proves the manager ran no trials outside the ledger. That residual gap closes only when registration precedes running, under forward pre-registration.

The oracle problem. The circuit reasons about committed data. It proves that the committed positions on the committed prices yield the committed metrics; it cannot certify that the prices are the true market history, or that the positions came from a genuine model rather than being authored directly. Data provenance rests on the source of the pinned dataset, a matter of committed reference and reputation, not of the proof.

Verifier-side checks. Certain quantities, the minimum backtest length and the protocol cost floor among them, are enforced by a conforming verifier against committed inputs rather than fully re-derived inside the circuit. We label these explicitly rather than present them as circuit invariants.

Economics are not modeled. The construction proves statistical properties of a return stream net of a disclosed cost model. It does not model capacity, market impact at scale, financing, or borrow costs — the frictions that determine whether a proven edge survives deployment. A credential is a statement about a strategy's honesty, not a guarantee of its future economics.

8. Conclusion

The anti-overfitting mathematics of systematic finance was built to tell skill from luck, and on honest inputs it succeeds. Its weakness was never the statistic but the fact that the party it constrains also supplies its inputs. By committing the trial ledger and recomputing the deflation in zero-knowledge, the correction stops being a number a manager can quietly soften and becomes a property of a credential that anyone can check in seconds, without ever seeing the strategy. Both schools of the literature can be evaluated this way; a strategy's honesty can be proven while its edge stays private; and a verification layer can, and should, publish its own failures. What remains — and it is the work we regard as most important — is to close the residual off-ledger gap by making commitment precede outcome: a forward-anchored, pre-registered track. That is the difference between honest history and a cryptographic live record, and it is the natural next layer of this construction.

Code and Data Availability

The independent recomputation script, the pinned dataset, and the disclosed fixture strategy are publicly available at mizan.market/research/reproduce. Every figure in §6 reproduces from them offline, in dependency-free Python, with no access to the prover, including the exact combinatorially-symmetric cross-validation over all $C(16,8) = 12,870$ splits. Credentials can be browsed in [the live registry](https://mizan.market/verify) and are independently verifiable at mizan.market/verify. The proving engine's source is not public; the reproducibility artifacts re-implement the scored arithmetic in the clear so the engine's honesty can be checked without trusting any MIZAN binary.

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Every numeric claim in the worked example is drawn from a live credential in the MIZAN registry and is independently re-verifiable via the public recompute script. The strategy is never revealed.